

YIN Haoteng 殷浩騰

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EDUCATION

- **Purdue University** West Lafayette, IN
• **Ph.D. in Machine Learning;** Department of Computer Science 2019 – Current
- **Peking University** Beijing, P.R. China
• **MPhil in Data Science;** Academy for Advanced Interdisciplinary Studies 2016 – 2019
- **Shandong University** Ji’nan, P.R. China
• **BEng in Computer Science and Technology;** Taishan College* (Honors School) 2012 – 2016
• **BEcon in Finance;** School of Economics 2014 – 2016
* Top-notch Undergraduate Training Program by Ministry of Education, P.R. China

RESEARCH EXPERIENCE

- **Purdue University** West Lafayette, IN
• *Graduate Researcher* Aug. 2019 - Current
 - Topics in Graph Representation Learning, Network Analysis & Computation, and Optimization
- **LIX, École Polytechnique & Inria** Palaiseau, France
• *Research Intern* Jun. 2018 - Sept. 2018
 - **Machine Learning Attacks to Location Privacy**
Research Project (supported by team Comète, EP), under the supervision of Prof. Catuscia Palamidessi.
Developed a model of adversary that uses machine learning algorithms to learn the trajectory pattern and the privacy mechanism of perturbed geospatial data gathered from location-based services, then performs privacy attacks on users’ location.
- **Center for Data Science, Peking University** Beijing, P.R. China
• *Graduate Research Assistant* Jan. 2017 - Aug. 2019
 - **Traffic Prediction with Deep Spatial Temporal Graph Neural Networks**
Research Project (supported by Deep Learning Lab, PKU), under the supervision of Prof. Zhanxing Zhu.
Designed a GNN-based general deep learning framework (STGCN) to coherently model the topology of road networks and forecast future traffic conditions of the network through space-time streaming data in the mid-and-long term. It has been widely used as a SoTA baseline for traffic studies (cited over 460 times).
- **School of Computer Science and Technology, Shandong University** Ji’nan, P.R. China
• *Undergraduate Researcher* Jan. 2016 - May. 2017
 - **Hotspot Prediction Based on Temporal Trajectory and Social Attributes**
Research Project (supported by Database Lab, SDU), under the supervision of Prof. Yang Liu.
Proposed a location-vector embedding framework (Loc2vec) to predict geospatial hotspots in a certain area and explore their semantic meanings based on temporal trajectories that are collected from users’ check-ins in location-based social networks.

PUBLICATIONS & MANUSCRIPTS

- **Haoteng Yin**, Yanbang Wang, Pan Li. Revisiting graph neural networks and distance encoding from a practical view. *The Fifth International Workshop on Deep Learning on Graphs (DLG-AAAI’21)*
- Bing Yu*, **Haoteng Yin***, Zhanxing Zhu. ST-UNet: A Spatio-Temporal U-Net for Graph-structured Time Series Modeling. *arXiv preprint arXiv:1903.05631* (* equal contribution)
- Bing Yu*, **Haoteng Yin***, Zhanxing Zhu. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *The 27th International Joint Conference on Artificial Intelligence (IJCAI’18)*. Main track. (* equal contribution)
- **YIN Haoteng** and LIU Yang. Semantic analysis of spatial temporal trajectory in LBSNs. (in Chinese) *SCIENTIA SINICA Informationis* 47(8), 1051-1065(2017). DOI: 10.1360/N112016-00310 (**Best Paper Award**)

RESEARCH INTERESTS & SKILLS

- **Interests:** Machine Learning and Geometric Deep Learning.
Topics in Optimization & Numerical Computation; Structured Deep Models (graph representation learning, spatiotemporal analysis); Privacy & Security (location privacy, robustness & generalization).
- **Languages:** Chinese (native), English (professional working proficiency, OEPT certified)
- **Programming:** Proficient in Python, Java, PyTorch, TensorFlow, L^AT_EX, Git
Experienced in C/C++, Julia, R, MATLAB

SELECTED HONORS & AWARDS

- Academic Excellence Scholarship, Peking University (~8%) *Oct. 2016, 2017, 2018*
- The 35th International Conference on Machine Learning (ICML'18) Student Volunteer Award *May. 2018*
- The 2nd DiDi-IEEE Elite Forum Extraordinary Potential Award (2/39) *May. 2018*
- École Polytechnique International Student (Intern) Fellowship *Mar. 2018*
- The Most Technical Difficulty Award (Team Leader), Schlumberger HackPKU *May. 2017*
- Guanghua-Yintai PKU Social Innovation DO Camp Scholarship (€1,600) *Jan. 2017*
- The 5th National Conference of Social Media Processing the Best Paper Award (1/25) *Oct. 2016*
- The Best Ideas for Banking Business, AK BARS Bank *Jul. 2016*
- The Best Show Prize for the 3rd Taishan College Academic Forum *May. 2016*
- The 3rd Prize Excellent Student Scholarship, Shandong University *Dec. 2013, 2014, 2015*
- Taishan College International Study Scholarship (¥33,000) *Jun. 2015*
- The 2nd Prize for the 1st Taishan College Academic Forum *Mar. 2014*
- The Provincial 2nd Prize for the China Undergraduate Mathematical Contest in Modeling *Dec. 2013*
- The 2nd Prize Excellent Student Scholarship, Shandong University *Dec. 2012*

PROFESSIONAL SERVICE

- Graduate Teaching Assistant - CS242, CS38003, CS490-LDA, CS590-DE I&II *Aug. 2019 -*
- Journal Reviewer *Dec. 2018 -*
 - ACM Transactions on Spatial Algorithms and Systems
 - IEEE Transactions on Big Data, Cybernetics, and Intelligent Transportation Systems
- Student Volunteer - The International Machine Learning Society *Jul. 2018*